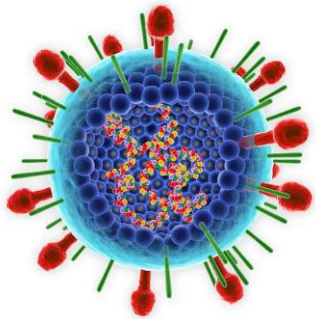


The Belgian National Survey on RSV pediatric burden of disease

Preliminary results

St Valentin's Symposium February 2025



Blumental S, De Mot L, Vandromme M, Lafort Y, Raes M
and the Belgian Pediatric RSV Study Group and the
BELSARI-net research group



Setting



→ **Creation of the RSVPED study**, run in parallel to SARI survey, bringing complementary pediatric data

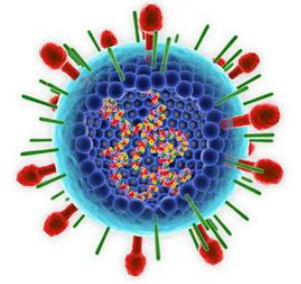
+ **Update of SARI protocol** to better catch pediatric cases (from November 2023)

→ **Pooled analysis of both datasets at the end**

Decision to build a **broad surveillance network** to better **assess the RSV pediatric burden of disease in Belgium (first in hospitals setting)** and to **monitor impact of preventive tools** once implemented

y infections in

Method

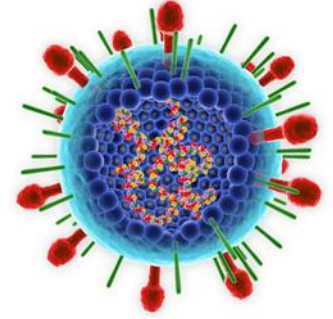


- ▶ **New common definition for pediatric severe acute respiratory infections** and a **new questionnaire** adapted to pediatric features, almost similar between **both studies**

- ▶ **SARI survey**
 - Prospective inclusion of all cases hospitalized for severe acute respiratory infections matching the case definition
 - Sample sent to the reference Lab, identification of respiratory viruses by pcr
 - Weekly clinical data report by participating centers (pseudonymized data)

- ▶ **RSVPED study**
 - Active recruitment of all centers in Belgium outside the SARI network
 - Inclusion of **all proven RSV** cases meeting the case definition and hospitalized during the enlarged epidemic period (October to March)
 - Retrospective **clinical data collection** by local team/data nurse (short or long CRF) (pseudonymized data for **season 2023-2024**)
 - Inclusion of **possible cases** for centers not testing all children hospitalized with bronchiolitis
 - Preliminary **aggregated data** obtained for the **3 first months of the current season**

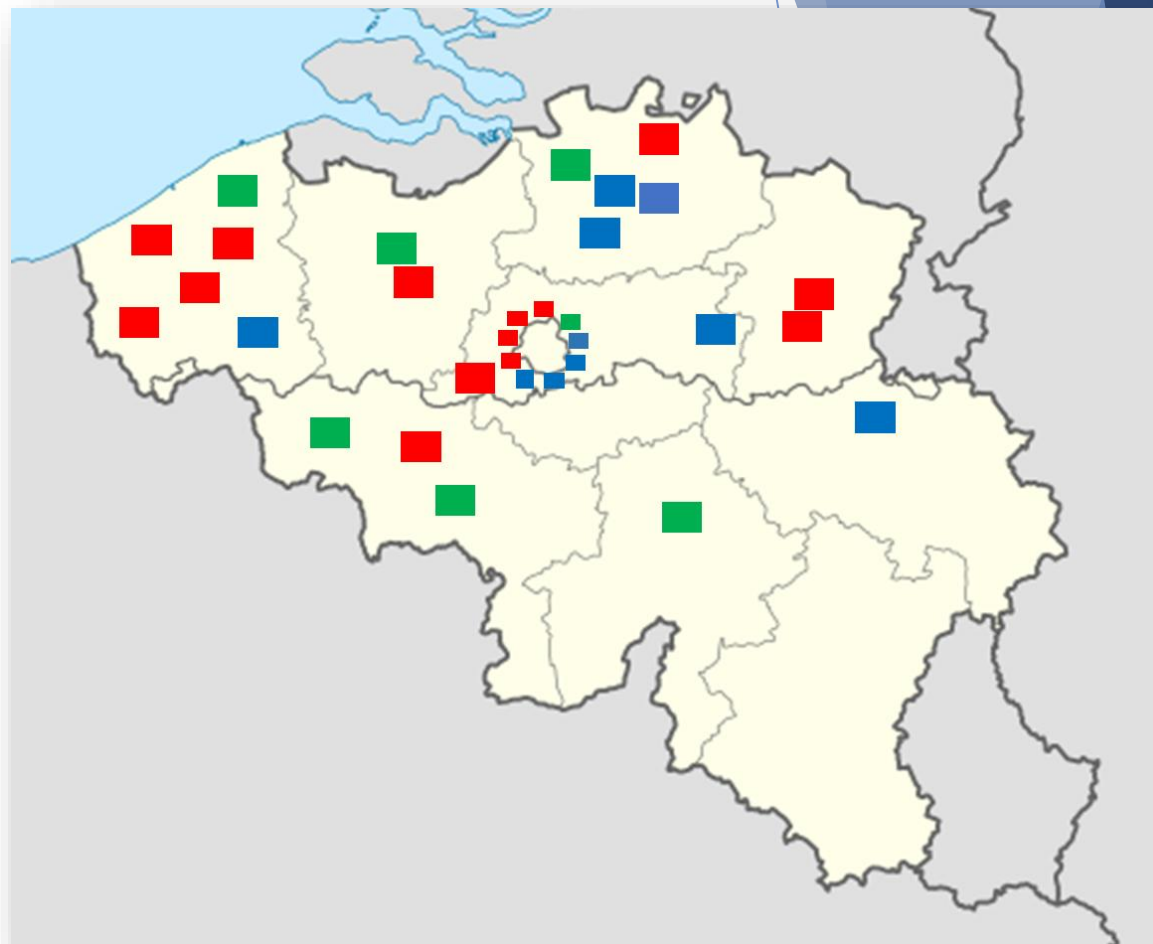
Timing



- ▶ **3 years of surveillance**
 - 2023-2024 (pre-prevention period);
 - 2024-2025 (first year of prevention implementation, one tool reimbursed),
 - Future 2025-2026 (second year of prevention, two tools reimbursed)
- ▶ **Merged of both datasets after each season and pooled analysis for common items**

31 participating sites all over the country

- ▶ **RSVPED** : 24 centers
 - 14 centers from which incidence data from both seasons are available (paired data)
 - 9 centers partial data collection
 - And many more coming...!!
- ▶ **SARI** : 7 additional centers



UZ Brussels, UZ Gent, AZ Sint Jan Brugge, Centre Chwapi Tournai, CHU Namur-Godinnes, Grand Hôpital de Charleroi, ZAS ziekenhuizen, CHU St Pierre, CHU Tivoli, AZ Klina, AZ Turnhout, AZ Delta Roeselare, Jessa Ziekenhuis, Clinique St Jean, AZ St Maria Halle, AZ Jan Palfijn Gent, AZ Groeninge, Jan Yperman, Hôpital Delta Chirec, Hôpital St Anne-St Remy Chirec, Hôpital de Braine-l'alleul, HUDERF, OLV Wareghem, CHC Mont Leggia, AZ Diest, Hôpital Iris Sud Ixelles, Vitaz, Cliniques de L'Europe, Cliniques Universitaires St Luc, Imelda Ziekenhuis

RSV: high pediatric burden of disease season 2023-2024

▶ RSVPED

- **2064 cases** included from **16 centers** (even under estimated!)
- 49% < 6 months old, 90% < 2 years old
- Median length of stay :5,4 days
- Use of respiratory support in 61% of the cohort (67% in infants<6m)
- Antibiotics consumption in 34% of children (26% in infants <6months)
- 165 ICU admissions recorded from 16 centers



Based on Belgian epidemiology in 2023-2024 (SARI): how many hospitalizations could hypothetically be prevented if eligible patients receive Nirsevimab?

Extrapolation of the benefits expectable based on:

- The **estimated incidence** for RSV hospitalizations in children 0-4y in 2023-2024 :
1049.8 per 100.000 inhabitants <5y old
- The **proportion of admitted children** that would have been **eligible for immunization**
among all hospitalized cases (before prevention available)
66% (newborns 29%; catch up 37%).
- The **percentage of uptake** of the preventive tool: ?? **85% maternity, 50% catch up**
- The **efficacy of the preventive tool** to prevent hospitalization (literature data) : **88%**

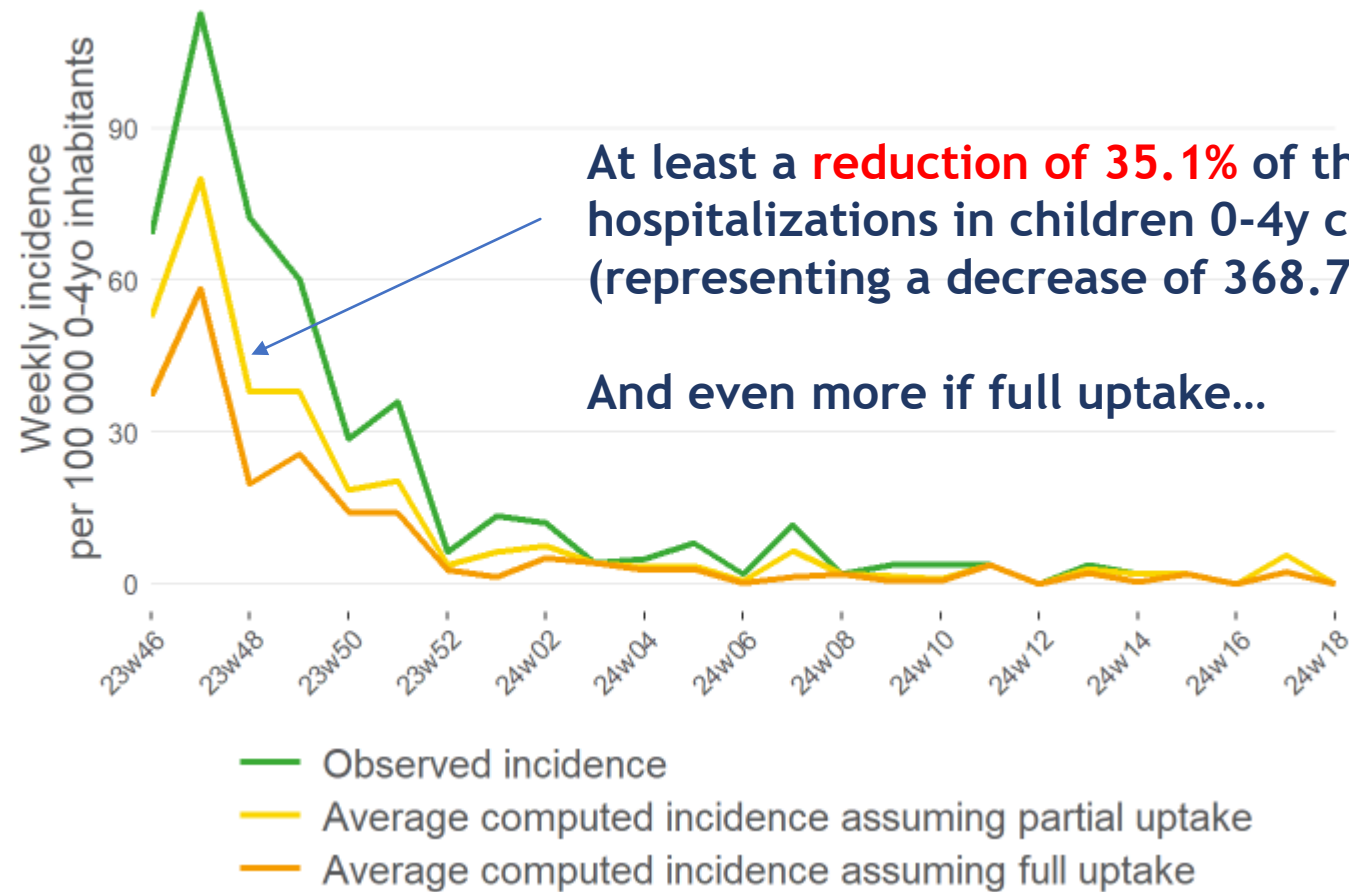
Uptake	Newborn	Catch-up
Low	0.75	0.3
Medium	0.85	0.5
High	0.9	0.6

X

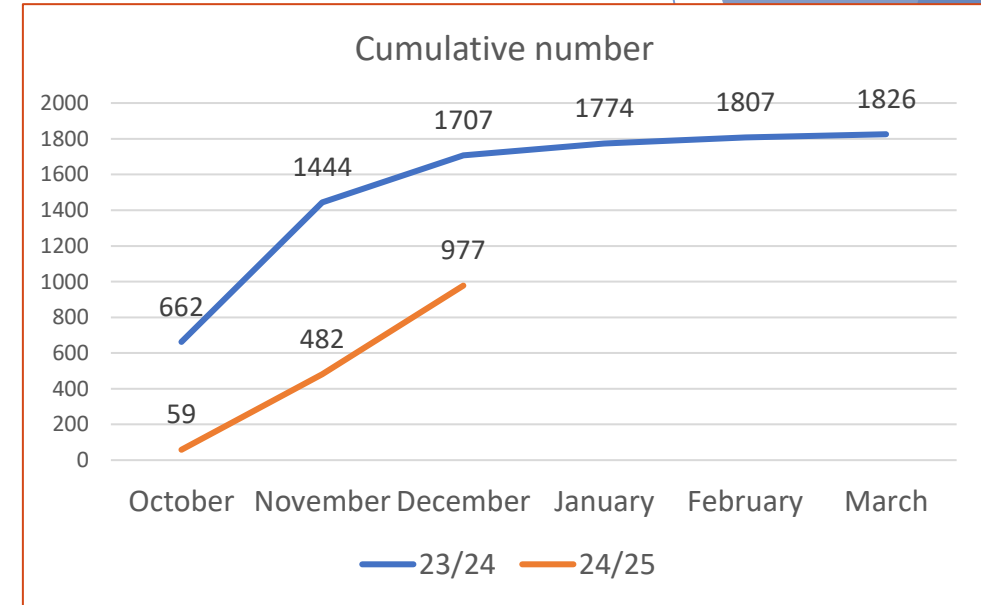
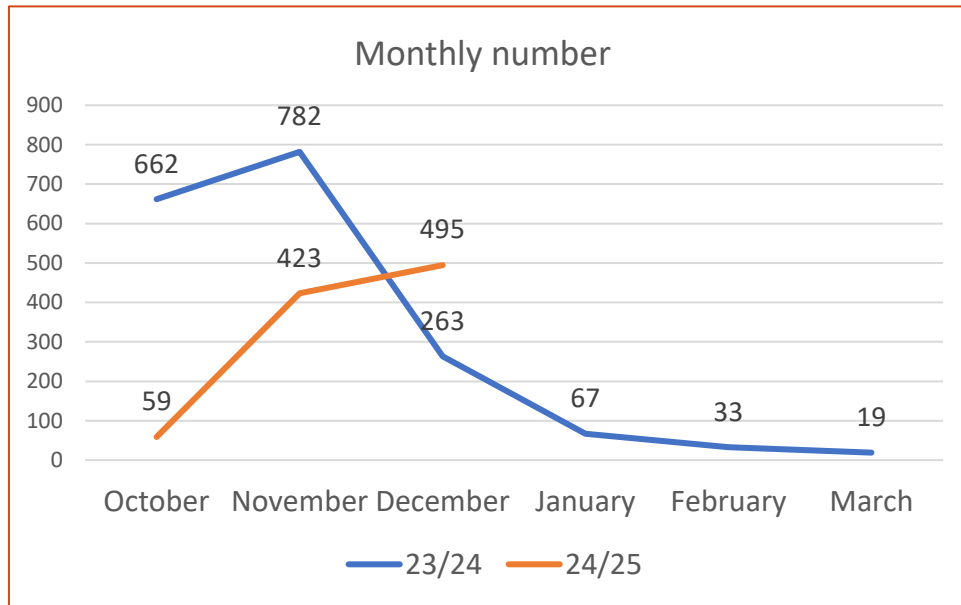
Efficacy	
Lower bound 95% CI	0.85
Average efficacy	0.88
Upper bound 95% CI	0.91



Based on Belgian epidemiology in 2023-2024, hypothetical impact of Nirsevimab on pediatric hospitalization using a medium uptake and average efficacy

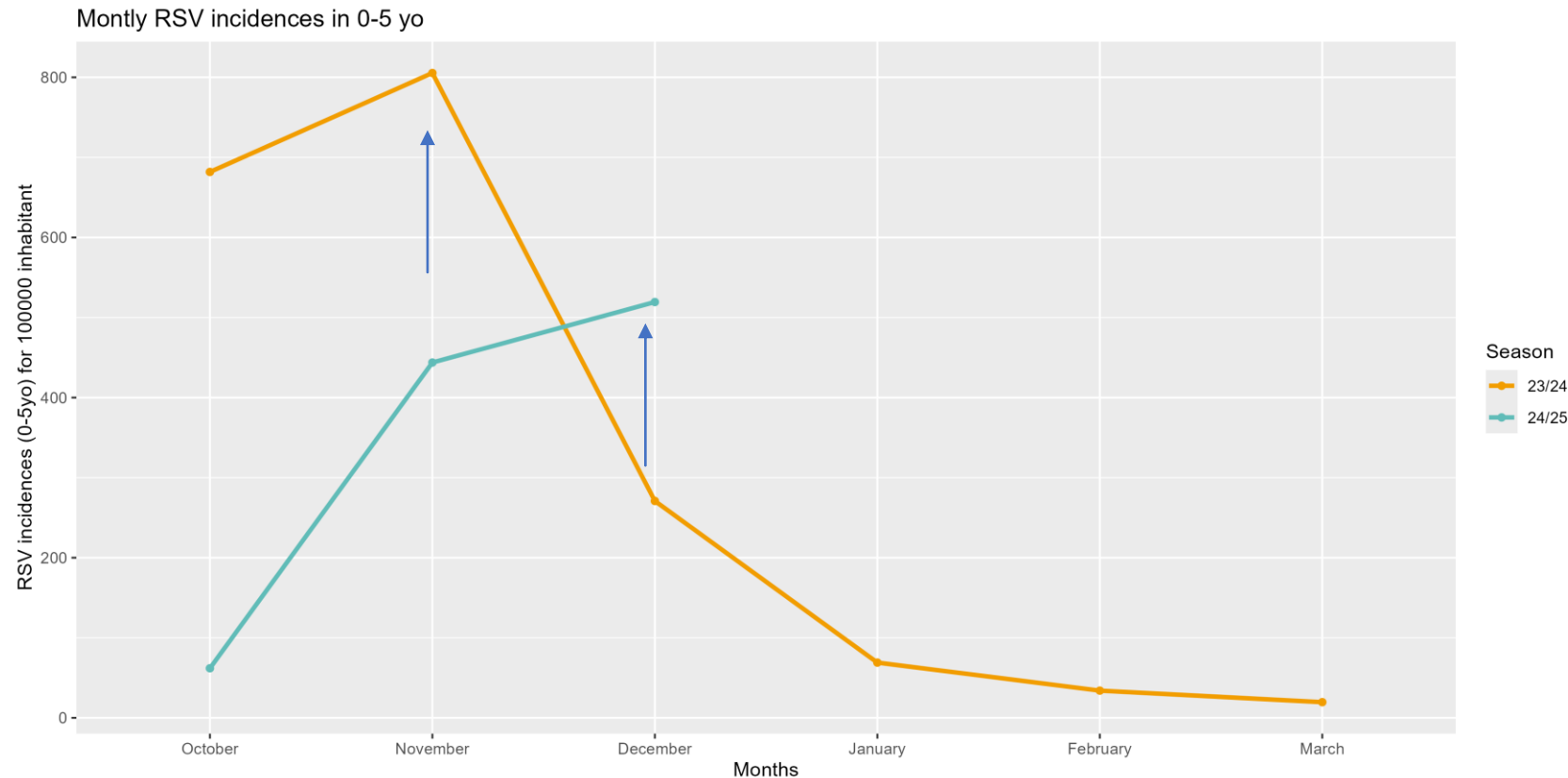


Number of children 0-4 y old hospitalized for RSV in 2024-2025 (first year prevention) compared to 2023-2024 (RSVPED)



- ❖ Between October and December: much less cases hospitalized this season compared to the previous one (! Actual season ongoing)
- ❖ Shift in seasonality: later this year (even more cases last season)

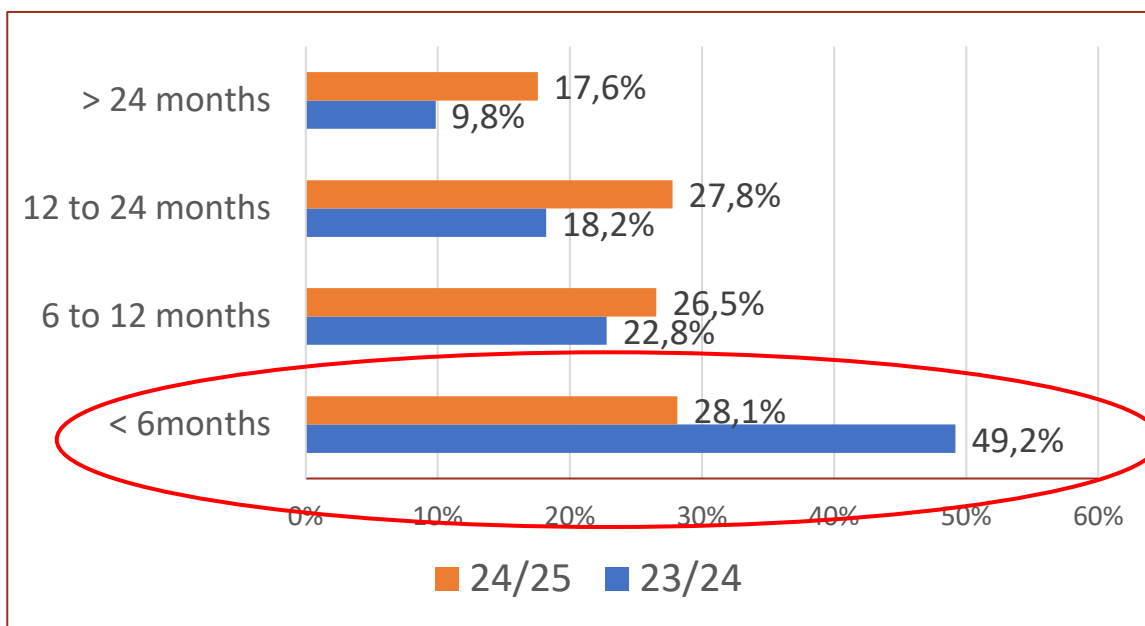
Incidence patterns of RSV hospitalizations in children 0-4 y old in 2024-2025 and 2023-2024 (RSVPED)



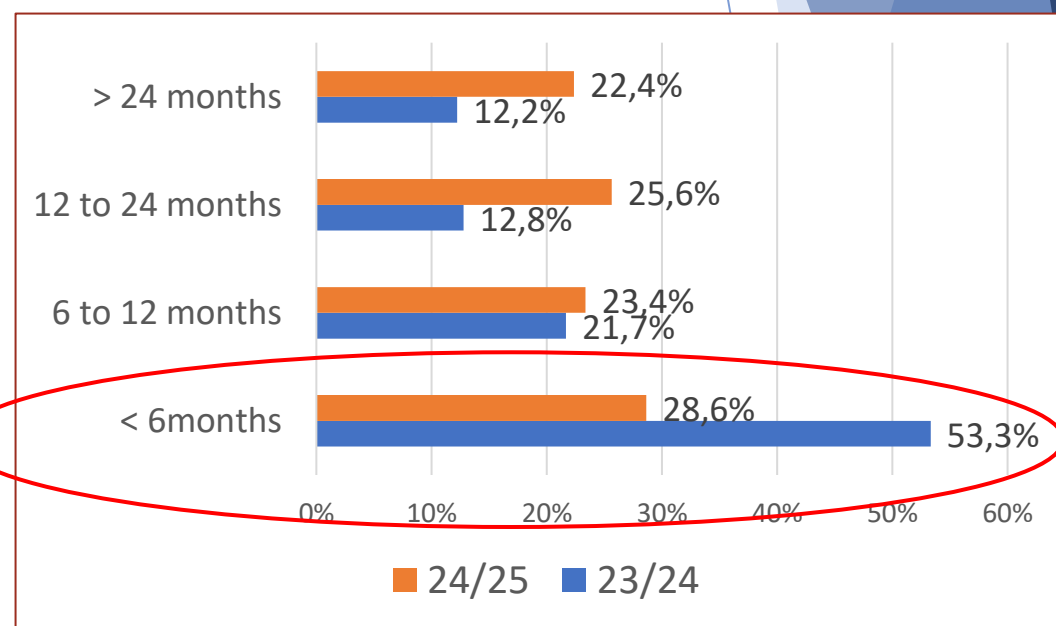
- ❖ **Incidence at peak level:** 450/100000 inhabitants 0-4y in 2024-25 versus 800/100000 inhabitants 0-4y in 2023-24

Age distribution among hospitalized cases in both seasons (RSVPED and SARI)

RSVPed

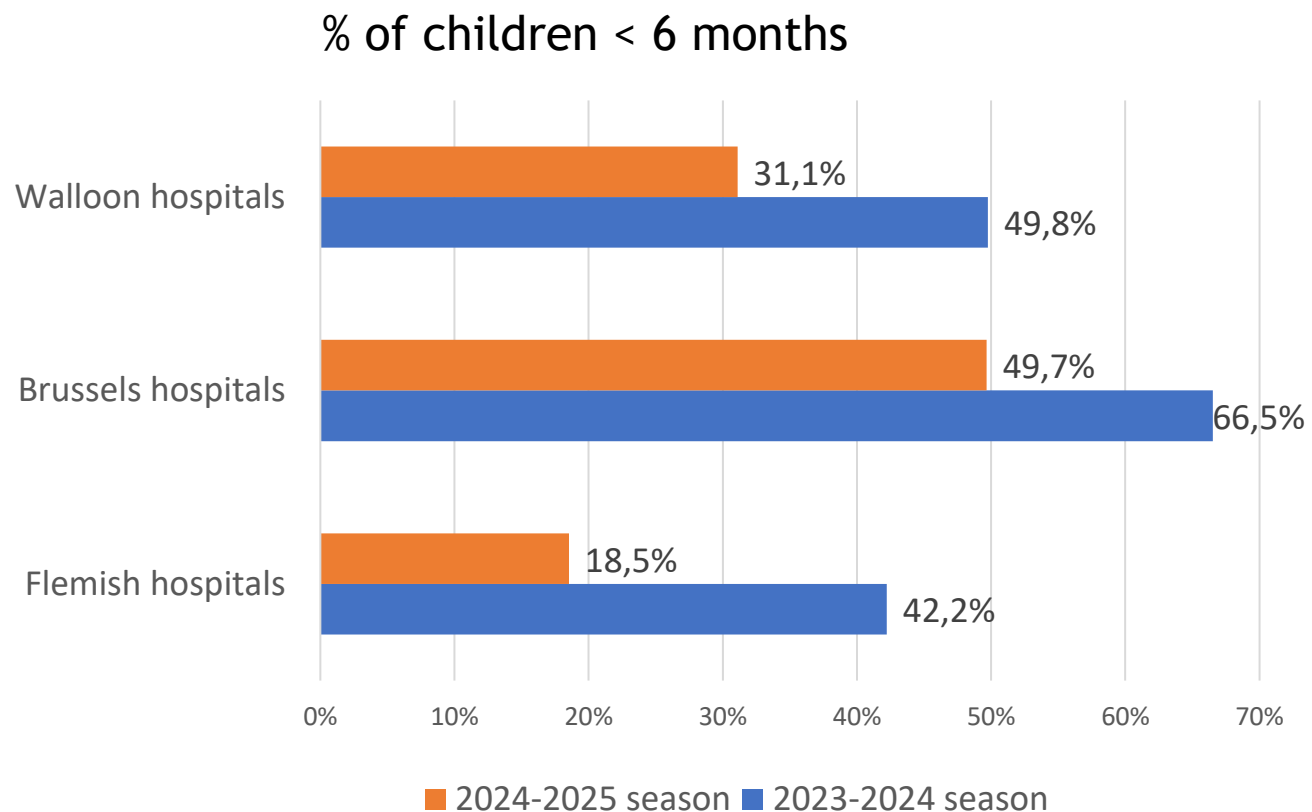


SARI



Age shift: <6 months proportionally less represented than last season
Same results in both studies

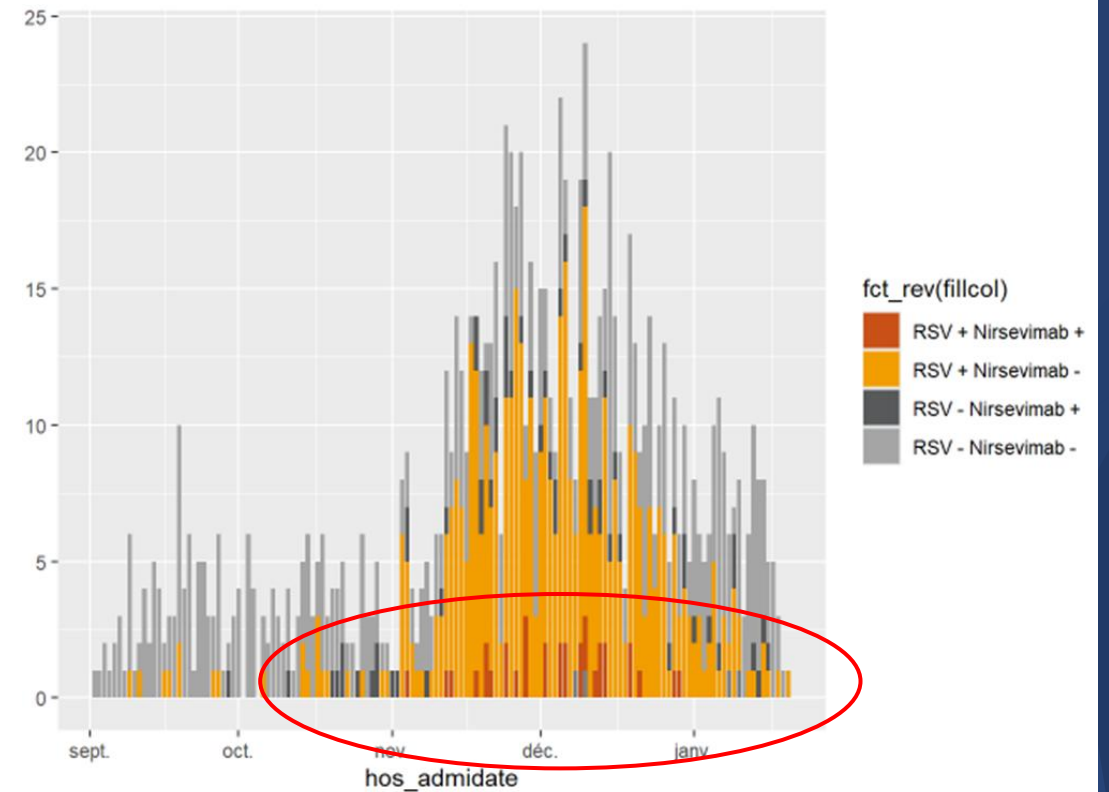
Age shift present in all three regions (pooled data)



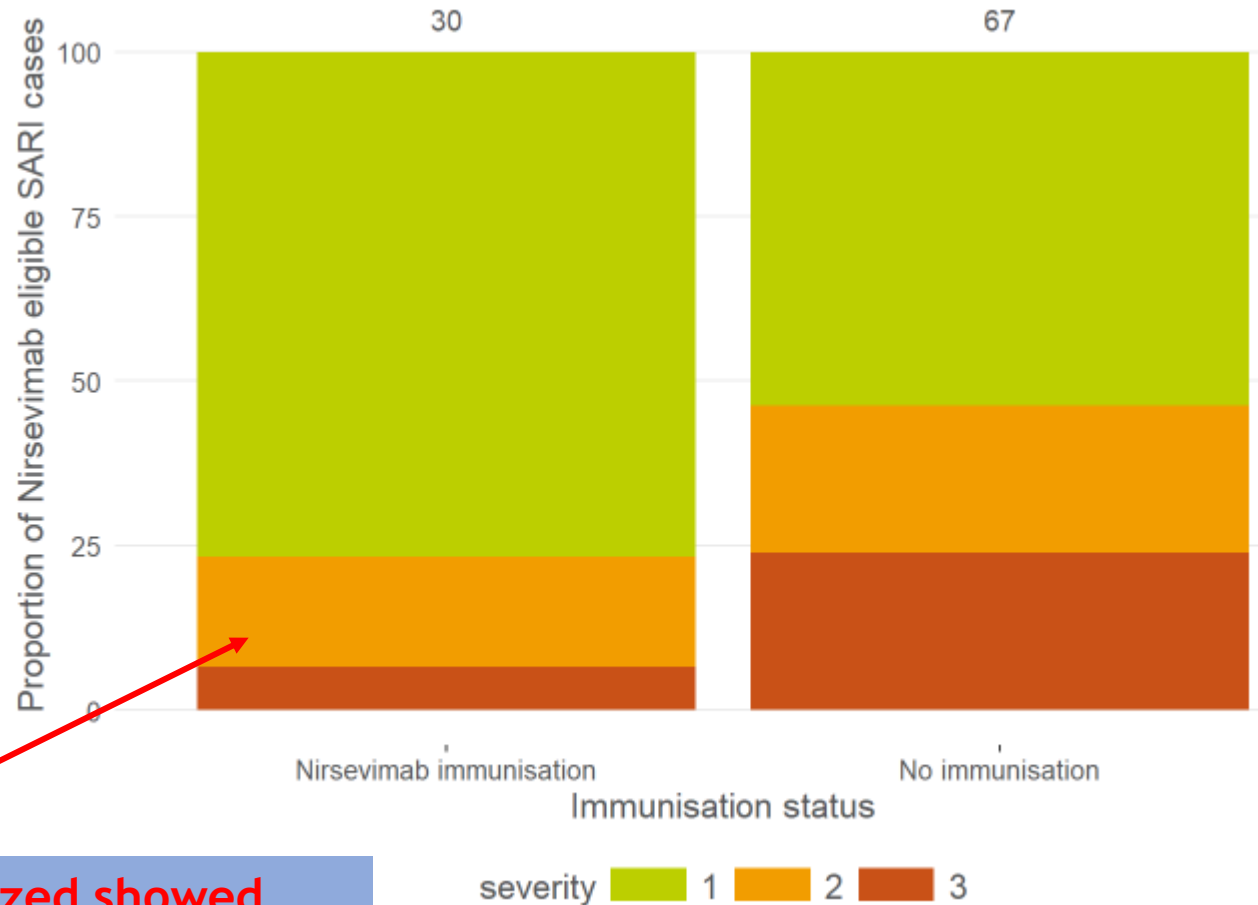
Highest difference seen in Flanders and among the <6 months olds

Immunization with Nirsevimab among hospitalized infants < 6 months (season 2024-2025)

- ▶ 33/129 cases in SARI (**25,6%**)
(14 catch up program, 18 newborns)
- ▶ 66/263 cases in RSVPed (**25,1%**)
(+ 7 with maternal vaccine)
- ▶ **This specific group :**
 - represents **6 to 7% of the whole cohort** of RSV hospitalized children
 - has **no more comorbidity** (3% vs 8% in all <6m))
 - has **no more prematurity** (1% vs 5,5% in all <6m)



Severity of infections in infants eligible for Nirsevimab according to immunization status (SARI, 2024-2025)



3 grades of severity
according to use of
respiratory support

Lowest severity (1)

None

Mid-severity (2)

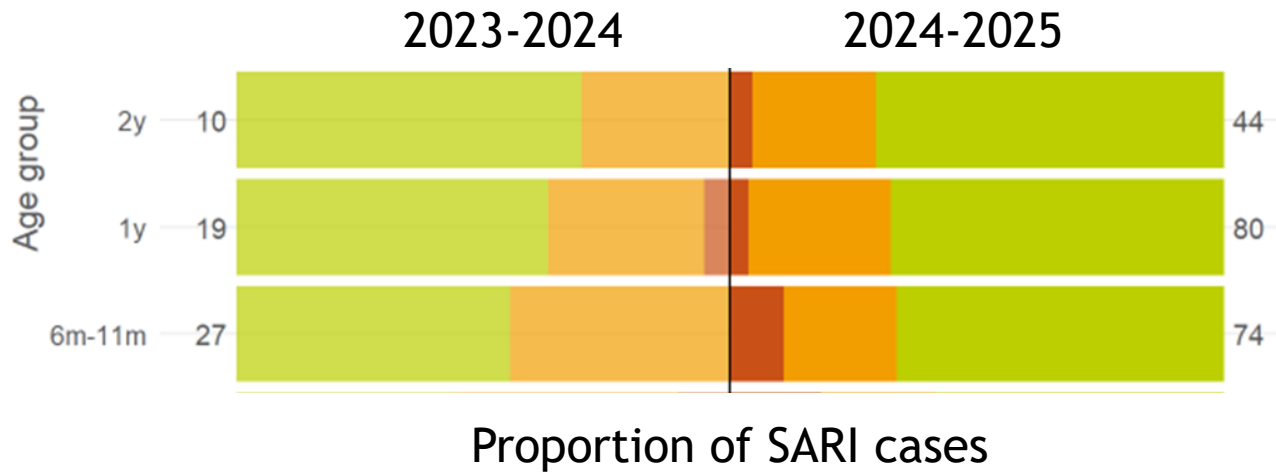
Non-invasive ventilation

Highest severity (3) invasive
ventilation and ICU admission

**Infants immunized showed
less severe infections!**

Severity of infection, older age groups

(% of cases by age groups, < 2y old, SARI data 2024-2025)



3 grades of severity according to use of respiratory support

Lowest severity (1)

None

Mid-severity (2)

Non-invasive ventilation

Highest severity (3)

invasive ventilation and ICU admission



Quite similar pattern between both seasons

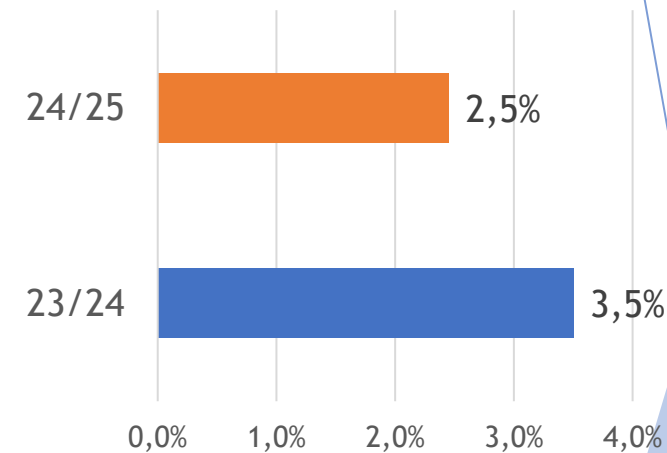
To be analyzed yet as severity markers:

length of stay, use of antibiotics, need for nutritional support, rate of pulmonary/neurological complications, nb of transfers to ICU, details of ICU stays...

Severity of infections/burden of disease ICU admissions (both studies)

- Reduction of ICU admission in RSVPed study
(! second season still ongoing)

RSVPED	2023-2024 season	2024-2025 season
Total hospitalized	1484	814
Admitted to ICU	52	20
% admitted to ICU	3.5%	2.5%



- No difference seen in the SARI survey or when both studies pooled
- Specific analysis in tertiary centers ongoing

What was the uptake for this first year?

If all eligible patients would have received Nirsevimab this season (100% uptake) : how many extra hospitalizations could have been prevented? (Oct 2024-Janv 2025, SARI data)

immunisation_group	Nirsevimab	No_Nirsevimab	total_number	hospitalizations
All patients	32	113	145	22.06897
catch-up	14	72	86	16.27907
newborn	18	41	59	30.50847

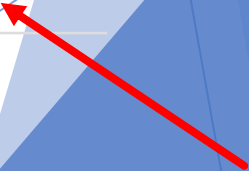
3/4 of infants <6m hospitalized non immunized..

113 x

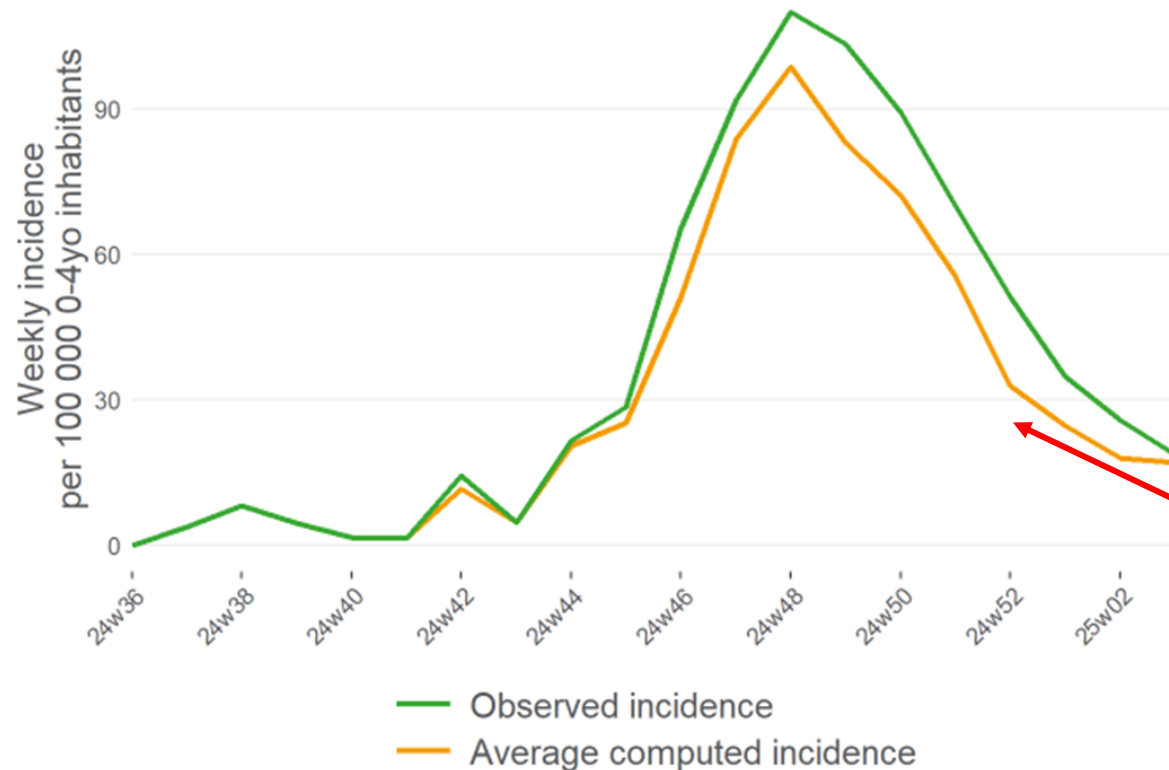
Efficacy	
Low CI efficacy	0.85
Average efficacy	0.88
High CI efficacy	0.91

=

Efficacy	Number of patients	Proportion of patients
Low CI efficacy	96	20.7
Average efficacy	100	21.6
High CI efficacy	103	22.2



If all eligible patients would have received Nirsevimab (100% uptake) this season (Oct 2024-Janv 2025, SARI data)



Incidence extrapolation:

Before removing patients:

749 per 100.000 0-4yo inhabitants

After removing patients:

618 per 100.000 0-4yo inhabitants

Difference of:

131 per 100.000 0-4yo inhabitants

→ **extra reduction of
17.4 % of the incidence observed**

Why was the catch-up uptake so low??

Acceptability of RSV prevention to Parents and Health Care Workers

Blumental S and Grossman D, Belgian Journal of Pediatrics Decembre 2024

- ▶ Multicenter opinion survey among young parents and antenatal caregivers in Brussels

→ **New acceptability study in preparation**

- ▶ Broader recruitment of parents:
Chirec Network, UZ Ghent, Jessa ziekenhuis + ONE, K&G and Kaleido
- ▶ + Assessment of opinion of pediatricians, gynecologists and midwives all over the country
- ▶ Goals: identify expectancies toward prevention
+ strengths and caveats of the current campaign
- ▶ The COVID-19 pandemic reduced the confidence of 24% of parents in vaccination

Take home messages

Promising results but we can do better!



- ▶ From Oct to end December: **reduction of RSV pediatric hospitalizations** in 2024-2025 (first season with RSV prevention) compared to 2023-2024
- ▶ **Highest impact** seen among the **< 6months old** (in terms of number of cases and severity)
- ▶ **Sub-optimal uptake** of the preventive tool, especially in the catch-up cohort



Efforts should be done to:

- Identify **expectancies** and feelings of **parents and health care workers** about RSV prevention
 - Improve **communication** about the campaign through various channels
 - Alleviate **administrative workload** linked to the preventive campaign
-
- ▶ Contribution of **every Belgian center** (whatever its size, its region or its budget) is **important to monitor prevention impact on a national scale**

Acknowledgements



- ▶ The team surveillance of respiratory infections (EID), Sciensano
Laurane De Mot, Mathil Vandromme, Yves lafort
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